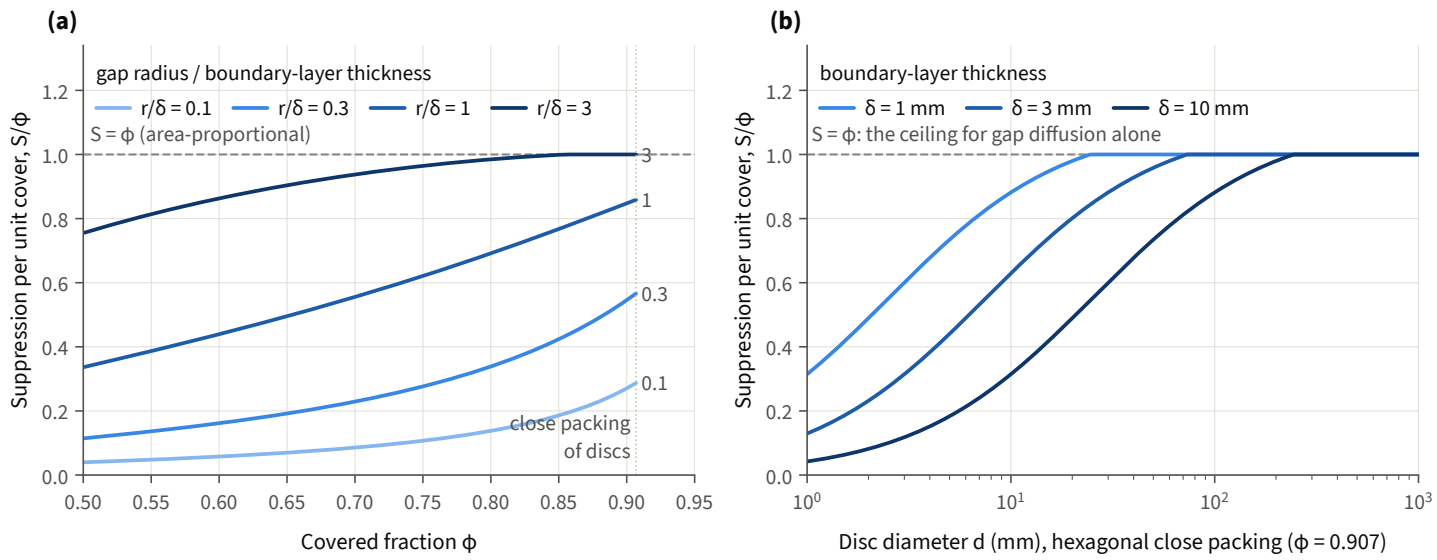
(a)
gap radius / boundary-layer thickness
r/δ = 0.1
r/δ = 0.3
r/δ = 1
r/δ = 3
S = ϕ (area-proportional)
3
1
0.3
0.1
close packing
of discs
Suppression per unit cover, S/ϕ
Covered fraction ϕ
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
(b)
boundary-layer thickness
δ = 1 mm
δ = 3 mm
δ = 10 mm
S = ϕ: the ceiling for gap diffusion alone
Suppression per unit cover, S/ϕ
Disc diameter d (mm), hexagonal close packing (ϕ = 0.907)
10^0
10^1
10^2
10^3